

Books Data - XML, JSON, and HTML

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Introduction

This assignment loads book information from three different file formats (HTML, XML, and JSON) into R data frames and compares them.

Load Required Libraries

```
options(encoding = "UTF-8")
Sys.setlocale("LC_ALL", "en_US.UTF-8")
```

```
## [1] "en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8"
```

```
library(XML)
library(jsonlite)
```

Book Information

Three books on interdisciplinary thinking and consciousness:

1. Gödel, Escher, Bach: An Eternal Golden Braid - Douglas Hofstadter (1979)
2. The Mind's I - Douglas Hofstadter & Daniel Dennett (1981)
3. The Black Swan - Nassim Nicholas Taleb (2007) *## Load Data from HTML*

```
# Parse HTML file
html_doc <- htmlParse("books.html", encoding = "UTF-8")

# Extract table
html_df <- readHTMLTable(html_doc, stringsAsFactors = FALSE)[[1]]

# Set column names
colnames(html_df) <- c("Title", "Authors", "Year", "Pages", "Award")

# Convert data types
html_df$Year <- as.integer(html_df$Year)
html_df$Pages <- as.integer(html_df$Pages)

print("HTML Data Frame:")
```

```
## [1] "HTML Data Frame:"
```

```
print(html_df)
```

```
##                               Title
## 1 Gödel, Escher, Bach: An Eternal Golden Braid
## 2                               The Mind's I
## 3                               The Black Swan
##                               Authors Year Pages      Award
## 1 Douglas Hofstadter 1979    777 Pulitzer Prize
## 2 Douglas Hofstadter, Daniel Dennett 1981    501      None
## 3 Nassim Nicholas Taleb 2007    366      None
```

```
str(html_df)
```

```
## 'data.frame':    3 obs. of  5 variables:
## $ Title   : chr  "Gödel, Escher, Bach: An Eternal Golden Braid" "The Mind's I" "The Black Swan"
## $ Authors: chr  "Douglas Hofstadter" "Douglas Hofstadter, Daniel Dennett" "Nassim Nicholas Taleb"
## $ Year    : int  1979 1981 2007
## $ Pages   : int  777 501 366
## $ Award   : chr  "Pulitzer Prize" "None" "None"
```

Load Data from XML

```
# Parse XML file
xml_doc <- xmlParse("books.xml", encoding = "UTF-8")

# Extract individual fields
titles <- xpathSApply(xml_doc, "//book/title", xmlValue)
years <- as.integer(xpathSApply(xml_doc, "//book/year", xmlValue))
pages <- as.integer(xpathSApply(xml_doc, "//book/pages", xmlValue))
awards <- xpathSApply(xml_doc, "//book/award", xmlValue)

# Handle multiple authors - collapse into single string
authors <- sapply(getNodeSet(xml_doc, "//book"), function(book) {
  author_nodes <- getNodeSet(book, ".//author")
  paste(sapply(author_nodes, xmlValue), collapse=", ")
})

# Create data frame
xml_df <- data.frame(
  Title = titles,
  Authors = authors,
  Year = years,
  Pages = pages,
  Award = awards,
  stringsAsFactors = FALSE
)

print("XML Data Frame:")
```

```
## [1] "XML Data Frame:"
```

```
print(xml_df)
```

```
##                               Title
## 1 Gödel, Escher, Bach: An Eternal Golden Braid
## 2                               The Mind's I
## 3                               The Black Swan
##                               Authors Year Pages      Award
## 1 Douglas Hofstadter 1979    777 Pulitzer Prize
## 2 Douglas Hofstadter, Daniel Dennett 1981    501      None
## 3 Nassim Nicholas Taleb 2007    366      None
```

```
str(xml_df)
```

```
## 'data.frame':    3 obs. of  5 variables:
## $ Title   : chr  "Gödel, Escher, Bach: An Eternal Golden Braid" "The Mind's I" "The Black Swan"
## $ Authors: chr  "Douglas Hofstadter" "Douglas Hofstadter, Daniel Dennett" "Nassim Nicholas Taleb"
## $ Year    : int  1979 1981 2007
## $ Pages   : int  777 501 366
## $ Award   : chr  "Pulitzer Prize" "None" "None"
```

Load Data from JSON

```
# Read JSON file
json_data <- fromJSON("books.json")

# Extract books array
json_df <- json_data$books

# Handle authors array - collapse to single string
json_df$Authors <- sapply(json_df$authors, paste, collapse=", ")

# Select and reorder columns
json_df <- json_df[, c("title", "Authors", "year", "pages", "award")]

# Rename columns to match other data frames
colnames(json_df) <- c("Title", "Authors", "Year", "Pages", "Award")

print("JSON Data Frame:")
```

```
## [1] "JSON Data Frame:"
```

```
print(json_df)
```

```
##                               Title
## 1 Gödel, Escher, Bach: An Eternal Golden Braid
## 2                               The Mind's I
## 3                               The Black Swan
```

```
##
##           Authors Year Pages      Award
## 1      Douglas Hofstadter 1979   777 Pulitzer Prize
## 2 Douglas Hofstadter, Daniel Dennett 1981   501      None
## 3      Nassim Nicholas Taleb 2007   366      None
```

```
str(json_df)
```

```
## 'data.frame':   3 obs. of  5 variables:
## $ Title : chr  "Gödel, Escher, Bach: An Eternal Golden Braid" "The Mind's I" "The Black Swan"
## $ Authors: chr  "Douglas Hofstadter" "Douglas Hofstadter, Daniel Dennett" "Nassim Nicholas Taleb"
## $ Year : int  1979 1981 2007
## $ Pages : int  777 501 366
## $ Award : chr  "Pulitzer Prize" "None" "None"
```

Compare Data Frames

```
# Test for identical data frames
cat("Are HTML and XML data frames identical?\n")
```

```
## Are HTML and XML data frames identical?
```

```
print(identical(html_df, xml_df))
```

```
## [1] TRUE
```

```
cat("\nAre HTML and JSON data frames identical?\n")
```

```
##
## Are HTML and JSON data frames identical?
```

```
print(identical(html_df, json_df))
```

```
## [1] TRUE
```

```
cat("\nAre XML and JSON data frames identical?\n")
```

```
##
## Are XML and JSON data frames identical?
```

```
print(identical(xml_df, json_df))
```

```
## [1] TRUE
```

```
# More flexible comparison
cat("\n--- More Flexible Comparison with all.equal() ---\n")
```

```
##
## --- More Flexible Comparison with all.equal() ---
```

```
cat("\nHTML vs XML:\n")
```

```
##  
## HTML vs XML:
```

```
print(all.equal(html_df, xml_df))
```

```
## [1] TRUE
```

```
cat("\nHTML vs JSON:\n")
```

```
##  
## HTML vs JSON:
```

```
print(all.equal(html_df, json_df))
```

```
## [1] TRUE
```

```
cat("\nXML vs JSON:\n")
```

```
##  
## XML vs JSON:
```

```
print(all.equal(xml_df, json_df))
```

```
## [1] TRUE
```

Display All Three Data Frames Side by Side

```
library(knitr)
```

```
cat("\n### HTML Data Frame\n")
```

```
##  
## ### HTML Data Frame
```

```
kable(html_df, caption = "Data loaded from HTML")
```

Table 1: Data loaded from HTML

Title	Authors	Year	Pages	Award
Gödel, Escher, Bach: An Eternal Golden Braid	Douglas Hofstadter	1979	777	Pulitzer Prize
The Mind's I	Douglas Hofstadter, Daniel Dennett	1981	501	None

Title	Authors	Year	Pages	Award
The Black Swan	Nassim Nicholas Taleb	2007	366	None

```
cat("\n### XML Data Frame\n")
```

```
##
```

```
## ### XML Data Frame
```

```
kable(xml_df, caption = "Data loaded from XML")
```

Table 2: Data loaded from XML

Title	Authors	Year	Pages	Award
Gödel, Escher, Bach: An Eternal Golden Braid	Douglas Hofstadter	1979	777	Pulitzer Prize
The Mind's I	Douglas Hofstadter, Daniel Dennett	1981	501	None
The Black Swan	Nassim Nicholas Taleb	2007	366	None

```
cat("\n### JSON Data Frame\n")
```

```
##
```

```
## ### JSON Data Frame
```

```
kable(json_df, caption = "Data loaded from JSON")
```

Table 3: Data loaded from JSON

Title	Authors	Year	Pages	Award
Gödel, Escher, Bach: An Eternal Golden Braid	Douglas Hofstadter	1979	777	Pulitzer Prize
The Mind's I	Douglas Hofstadter, Daniel Dennett	1981	501	None
The Black Swan	Nassim Nicholas Taleb	2007	366	None

Conclusion

Are the three data frames identical?

```
if (identical(html_df, xml_df) && identical(html_df, json_df)) {
  cat("YES - All three data frames are completely identical.\n")
} else {
  cat("NO - The three data frames are NOT identical.\n\n")
  cat("However, they contain the same information. Differences may be due to:\n")
  cat("- Different internal data structure representations\n")
  cat("- Row names or attributes varying by parsing method\n")
  cat("- Minor differences in how R stores the data internally\n\n")
  cat("Using all.equal() shows they are functionally equivalent.\n")
}
```

```
## YES - All three data frames are completely identical.
```

The three data frames contain the same information but may not be strictly identical due to differences in how each parsing method stores data internally. The `all.equal()` function provides a more flexible comparison that shows they are functionally equivalent.